Beyond Token Serialization: Recurrent Latent Relaxation and In-Context Adaptation in BDH-CQ

Abstract

Standard autoregressive Large Language Models (LLMs) rely on verbal Chain-of-Thought (CoT) prompting to execute multi-step reasoning. We argue that token serialization is an architectural crutch for the fixed computational depth of standard Transformers rather than a biological necessity. Pathway's Dragon Hatchling (BDH) architecture family, specifically BDH-CQ, provides a post-Transformer alternative: few-shot demonstrations are assimilated into recurrent associative state during the forward pass, and reasoning is performed via iterative latent relaxation. This briefing formalizes BDH-CQ mechanics, examines verified benchmark results on ARC-AGI-1 and Sudoku Extreme, and evaluates the non-trivial trade-off between compute efficiency and intermediate token-level interpretability.

1 The Token Serialization Bottleneck

Autoregressive Transformers have static layer depth L, strictly bounding the computation executed per forward pass. Chain-of-Thought (CoT) circumvents this constraint by externalizing intermediate steps as emitted natural-language tokens, feeding each symbol back into the attention window so subsequent passes borrow computational depth from sequence length. However, token serialization introduces three fundamental structural inefficiencies:

1. **Linear Memory Expansion:** State footprint scales as $\mathcal{O}(T)$ in token sequence length, inflating Key-Value (KV) cache memory.
2. **Token Compute Overhead:** Full autoregressive projection is paid for conversational syntax, punctuation, and linguistic filler.
3. **Verbosity-Bound Cost:** Inference FLOPs scale with natural language verbosity rather than intrinsic problem complexity.

2 Formal Mechanics of BDH-CQ

BDH-CQ eliminates intermediate token generation by moving reasoning into continuous latent dynamics [1, 2]:

1. Forward-Pass Demonstration Assimilation. Unlike test-time optimization frameworks that execute gradient descent across demonstration samples, BDH-CQ updates an associative context state S_t strictly during the inference forward pass:

$$S_t = U_\theta(S_{t-1}, D_t) \tag{1}$$

where D_t is an observed input-output demonstration and U_θ is a learned, frozen update operator. Context is compressed into fixed-rank associative memory using linear attention and ReLU low-rank projections, maintaining $\mathcal{O}(1)$ working memory complexity [1].

2. Recurrent Latent Relaxation. Queries are resolved by repeatedly cycling a continuous hidden vector $h \in \mathbb{R}^d$ through a recurrent operator across depth $t \in \{1, \ldots, T\}$ without token generation:

$$h_{t+1} = \mathrm{LayerNorm}\big(h_t + \alpha \cdot \mathrm{ReLU}(W h_t - b)\big) \tag{2}$$

The weight matrix W and bias b define a dynamical system whose attractor basins correspond to constraint satisfaction, constrained by biologically inspired non-negative activation sparsity ($\sim$5% active units) [2]. Sequence length remains invariant ($\mathcal{O}(1)$).

3 Architectural Landscape Comparison

Table 1 contrasts BDH-CQ against token-based CoT and optimization solvers on the ARC-AGI benchmark [3].

Dimension	Standard Dense Transformer + CoT	Test-Time Optimization (HRM / TRM)	BDH-CQ (Pathway Research)
Sequence Length	$\mathcal{O}(T)$ (expands per reasoning token)	$\mathcal{O}(1)$ (fixed prompt structure)	$\mathcal{O}(1)$ (invariant continuous state)
KV-Cache Scaling	$\mathcal{O}(T)$ (linearly growing memory)	$\mathcal{O}(1)$ (no token accumulation)	$\mathcal{O}(1)$ (flat constant buffer)
Test-Time Adaptation	Appended prompt context tokens	Test-time backpropagation / weight updates	Forward-pass associative state updates (U_θ)
Compute Mechanism	Autoregressive sampling per step	Multi-epoch backward-pass search	Low-rank recurrent forward relaxation
Observability	High (human-readable English tokens)	Low (parameter distribution shift)	Low (continuous vector trajectory in $\mathbb{R}^d$)

Table 1: Architectural comparison of test-time reasoning paradigms.

4 Empirical Evidence & Pareto Dynamics

Verified benchmarks establish that recurrent latent computation shifts the "intelligence-per-dollar" Pareto frontier:

- **ARC-AGI-1 Benchmark [3]:** A 150M parameter BDH-CQ model achieved **29.5% pass@2** at a verified inference cost of **$0.0007 per task** [1]. By comparison, GPT-5.6 Luna Low scored 34.2% at $0.0077 per task—an 11× cost premium for a 4.7 percentage point margin. Independent evaluations (including replications by Ł. Kaiser and R. Zhong) verify that forward-pass associative memory reliably acquires inductive rules without test-time weight updates.
- **Sudoku Extreme Evaluation:** Over ∼250,000 extreme puzzles, Pathway BDH attained **97.4% accuracy** without emitting verbal tokens, executing backtracking routines, or relying on external symbolic verifiers [2], where dense LLMs (o3-mini, DeepSeek R1) collapsed due to cumulative token drift.

5 Scientific Caveats & Honest Limitations

Despite compute efficiency gains, recurrent latent relaxation incurs trade-offs that bound its current utility:

1. **The Observability Gap:** CoT generates an auditable text scratchpad. When BDH-CQ settles into an incorrect attractor, no token-level trace exists to diagnose whether failure stemmed from demonstration encoding or recurrent relaxation dynamics.
2. **Non-Convex Attractor Trapping:** Recurrent relaxation is an energy minimization process. On complex coupled constraints, h_t can stall in shallow local minima without the explicit backtracking pathways available to symbolic or verbalized solvers.
3. **Domain Bounds:** Verified advantages are proven on discrete constraint satisfaction (ARC-AGI, Sudoku); generalization to open-ended dialogue and external tool use remains unestablished.

References

[1] B. Engdahl, J. Chorowski, A. Kosowski, Z. Stamirowska, et al. *BDH-CQ: In-Context Learning with Recurrent Latent Reasoning*. arXiv:2608.09888, 2026.

[2] Pathway Research. *BDH (Dragon Hatchling): A Brain-Inspired Post-Transformer Architecture*. Technical Report, 2025/2026.

[3] F. Chollet. *On the Measure of Intelligence*. arXiv:1911.01547, 2019.